

V Novel Mathematical Structures for String Theory, Extra D, and AdS/CFT



Functors of Actions Theories (FAT)
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App to Cosmos: Cosmic implications of FATs
 $G_{\mu\nu}[\partial g, \partial^2 g] \left\{ 1 + \frac{\delta F}{\delta \int d^D x \sqrt{-g} R(\partial^2 g)} \right\} = \kappa^2 T_{\mu\nu}[g]$

App to Strings: Flux Compactification Amplitudes

Analytical poly Λ CDM dynamics
Effective potentials from Calabi-Yau compactifications may realize poly Λ CDM dynamics

Standard Λ CDM + Modified Gravity and DE

Strings ?
CY-Potential
+ Compactification
Amplitudes ?

$L = (m, r, \Lambda) + (IDE, f(R), \text{Curvature, SVT})$

$H^2(t) = H_0^2 \sum_{s \in \{m, r, \Lambda, x, k, z, v\}} \Omega_{s0} a^{-\alpha_s}(t)$

$H_{CY}^2(t) = H_0^2 \Omega_{CY0} a^{-\alpha_{CY}}(t)$

A (D, τ, D_x) -manifold with NPCF of Nt-objects

- extra dimensions
- contaminated targets
- Applied to Cosmic Structures:
 - Large scale structure (Astronomical)
 - Small scale structure (Quantum)

String application: holographic correlators with scale-dependent behaviour from AdS/CFT

where $f_{\alpha}(\tau, \delta)$ is defined as the constant factor of targeted elementary particles which are contaminated by any natural contaminants appear at the level of their detection, such as non-targeted elementary particles, from other generalised spacetime regions, or composition of elementary particles, of even cosmic rays. Then, we can define the contaminated targeted functional as

$C^{(T)}(x) = \sum_{\alpha} D_{\alpha}^{(T)}(x) \left\{ 1 - \sum_{\alpha} f_{\alpha}(\tau, \delta) \right\} + \sum_{\alpha} \gamma_{\alpha} D_{\alpha}^{(T)}(x) f_{\alpha}(\tau, \delta) D_{\alpha}^{(T)}(x) \quad (3.48)$

This means that $\phi(\tau) = \phi_{\alpha} C^{(T)}(x)$. In this case we have that the NPCF for a quantum field, ϕ is analyzed as

$C_{\phi}(\tau, \delta_1, \dots, \delta_k) = \int_{\mathcal{M}} D_{\phi}(\delta_1) \dots D_{\phi}(\delta_k) C^{(T)}(\tau, \delta_1) \dots C^{(T)}(\tau, \delta_k) \mathcal{L}(\tau, \delta_1, \dots, \delta_k) \quad (3.50)$

Amplitudes ?

Universions

Advanced Manifold Metric Pairs

$ds^{U+L} = F_{\mu\nu}^{\nu} \tilde{F}^{\nu} \tilde{F}^{\nu} [c, f(c)] dc^{\mu} dc^{\mu} dc_{\nu}^{\mu} dc_{\nu}^{\mu} dc_{\nu}^{\mu}$

$c = [t, \vec{x}, \vec{P}, \vec{I}, \vec{S}]$

Geometrical, Topological, and Physical Applications

Strings APP: generalised geometries for flux compactifications and quantum field theory

Flux Compactification Amplitudes ?

$ds^2 = e^{A(y)} g_{\mu\nu} dx^{\mu} dx^{\nu} + e^{-A(y)} \tilde{g}_{mn} dy^m dy^n$

Advancing Tensor Theories

$\mathcal{T} : \mathcal{V} \rightarrow \mathbb{G}$

e.g. $T = T^{\mu\nu} dx^{\nu} \otimes (\partial_{\beta})^{1/2} \otimes \partial_{\mu}$

$T^{00} = 1$, where $i, j, k \in \{0\}$

$T^{11} = 0.5$, where $i, j, k \in \{1\}$

$T^{ijk} = 0$, $i, j, k \in \text{otherwise}$

$\Rightarrow$ Geometrical and Topological applications : non-linear, non-local worldsheet theories or fractional-brane dynamics

Amplitudes ?

Science Popularisation !

Universions

YouTube Analogue

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Xiè xiè
Ευχαριστώ
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CHINA
JULY 6-10,
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QR codes and links:

- A cosmic femtolensing Hawking radiation
- poly Λ CDM (in cosmic observations) (Talk)
- Functors of Actions Theories (FAT) (Cosmos APP)
- Quintessence
- Extra Dimensions in Observations
- Advanced Manifold-Metrics Pairs (AMMP)
- A temporal-spatial-entropic K curvature
- EinsteinKPy2Tex code
- Advanced Tensor Theories (ATT)

Diagram remodified from P.Ntelis & A.Morris '23 Ezquiaga & Zumalacárregui 18 and novel theories

Cosmological GRAvitational TheORy (COGRATORY)

General Relativity: unique theory of massless $g_{\mu\nu}$

Massive Gravity $m_g > 0$

Tensor $T_{\mu\nu}$

Additional Field

Vector V_{μ}

Scalar ϕ

Love-lock

Quint-essence

Brans-Dicke

f(R)

Hordenski

Gauss-Bonnet

Beyond Hordenski

Galleon

Galleon UEDM

KGB

C(X)

D(X)

E(X)

Break Assumptions

(Non)isotropy (Non)homogeneity

(Non)locality

(Non)causality

Curvature types (R, T)

Torsion (T)

Non-Riemannian

Mathly interesting

Advanced Manifold-Metric Pairs

Extra Dimensions

Strings/Branes

Advanced Tensor Theories

Fractional Calculus

Entropic Dimensions

Probabilistic Dimensions

Backreaction

Non-metricity (Q)

Combined (R, T, Q)

